

Hoare Logic

Models of Computation

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Introduction

Hoare Logic is a formalism relating the initial and terminal state of a program.

Invented in 1969 by Sir Tony Hoare, inspired by earlier work of Robert Floyd.

Still being researched.

General Idea

Hoare introduced "partial correctness specification" (Hoare Triple) for specifying the behaviour of a program:

$$\{P\} C \{Q\}$$

C is a command

P is a Precondition, describing the initial state

Q is a Postcondition, describing the final state

General Idea II

$$\{P\} C \{Q\}$$

- when C is executed in a state satisfying P
- if the execution of C terminates
- then the state in which C terminates satisfies Q

Conditions on program variables are written using mathematical notation together with logical operators. E.g.:

$$\{x = 1\} y := x + 1 \{y = 2\}$$

Note on Logical Variables

Imagine we did a multiplication as follows

$$\{ x > 0 \} x := x + x; y := x; \{ y = 2 * x \}$$

Does this work?

Better version:

$$\{ x_{log} = x \wedge x > 0 \} x := x + x; y := x; \{ y = 2 * x_{log} \}$$

What problems can arise?

Function Bodies - Example

```

1  fun squareOfSum(x: Int, y: Int): Int
2  // PRE: true                                     (P)
3  // POST: r = (x + y)2                             (Q)
4  {
5      var c = x * y
6      // MID: c = xy                                (M1)
7      val b = y * y
8      // MID: b = y2 ∧ c = xy                        (M2)
9      val a = x * x
10     // MID: a = x2 ∧ b = y2 ∧ c = xy              (M3)
11     c = c + c
12     // MID: a = x2 ∧ b = y2 ∧ c = 2xy            (M4)
13     return a + b + c
14 }
```

Hoare Logic

Hoare Logic is a deductive proof system for Hoare triples. It can be used:

As a semantics ("Axiomatic semantics")

To show properties of programs.

Hoare Logic

In order to do the above meaningfully, we need:

- The language WHILE and its operational semantics (we will use Big-Step)
- Language for defining state predicates and its semantics (“assertion language”)
- Formal interpretation of triples and a proof system for deriving them.

Assertion Language

Assertion language is an instance of first-order logic with equality, with values as integers, and assertions properties of WHILE states.

Assertion language:

$$\text{var} \in \text{Variables} ::= \text{id} \mid \text{id}_{\log}$$
$$t \in \text{Terms} ::= \text{var}$$
$$P, Q \in \text{Assert} ::= \text{true} \mid \text{false} \mid P \wedge Q \mid P \vee Q \mid P \Rightarrow Q \mid \\ \forall \text{id}. P \mid \exists \text{id}. P \mid t_1 = t_2$$

Note, $\neg P$ can be represented as $P \Rightarrow \text{false}$.

Semantics I: Terms

semantics of a term t in a state s : $\llbracket t \rrbracket(s)$.

$$\llbracket \cdot \rrbracket: ? \times ? \rightarrow ?$$

$$\llbracket v \rrbracket(s) \triangleq s(v)$$

Semantics II: Assertions (a)

$\llbracket P \rrbracket$: the set of states that satisfy the assertion P .

$\llbracket \cdot \rrbracket$: $? \rightarrow ?$

$\llbracket false \rrbracket \triangleq \emptyset$

$\llbracket true \rrbracket \triangleq ?$

$\llbracket P \vee Q \rrbracket \triangleq \llbracket P \rrbracket \cup \llbracket Q \rrbracket$

$\llbracket P \wedge Q \rrbracket \triangleq ?$

$\llbracket P \Rightarrow Q \rrbracket \triangleq ?$

Semantics III: Assertions (b)

$$\llbracket t_1 = t_2 \rrbracket \triangleq \{s \in \text{State} \mid \llbracket t_1 \rrbracket(s) = \llbracket t_2 \rrbracket(s)\}$$

$$\llbracket \forall v. P \rrbracket \triangleq \{s \in \text{State} \mid \forall n \in \mathbb{N}. s[v \mapsto n] \in \llbracket P \rrbracket\}$$

$$\llbracket \exists v. P \rrbracket \triangleq ?$$

Semantics V: Semantics of Hoare Logic

Partial Correctness

Definition For any assertions P , Q , and command C :

$$\models \{P\} C \{Q\} \triangleq \forall s, s' \in \text{State}. [s \in \llbracket P \rrbracket \wedge \langle C, s \rangle \Downarrow s' \implies s' \in \llbracket Q \rrbracket]$$

Derivation Rules for Hoare Logic I

conspicuous absence of skip

$$\frac{}{\vdash \{P[E/X]\} x := E \{P\}} \quad H - \text{ASSIGN}$$

$$\frac{\vdash \{P\} C_1 \{Q\} \quad \vdash \{Q\} C_2 \{R\}}{\vdash \{P\} C_1; C_2 \{R\}} \quad H - \text{SEQ}$$

Derivation Rules for Hoare Logic

$$\frac{\vdash \{P \wedge B\} C_1 \{Q\} \quad \vdash \{P \wedge \neg B\} C_2 \{Q\}}{\vdash \{P\} \text{if } B \text{ then } C_1 \text{ else } C_2 \{Q\}} \quad H - IF$$

$$\frac{\vdash \{P \wedge B\} C \{P\}}{\vdash \{P\} \text{while } B \text{ do } C \{P \wedge \neg B\}} \quad H - WHILE$$

$$\frac{\vdash_L P_S \Rightarrow P_W \quad \vdash \{P_W\} C \{Q_S\} \quad \vdash_L Q_S \Rightarrow Q_W}{\vdash \{P_S\} C \{Q_W\}} \quad H - CNSQ$$

Soundness and Completeness

So now we have a (syntactic) method for deriving Hoare triples, and a formal definition of their meaning.

What we want to do is *relate* them.

Desired properties:

- (1) Soundness: any triple that can be derived syntactically holds semantically,
- (2) Completeness: and that any triple that holds semantically is syntactically derivable.

Do we have these? What about Decidability?

Soundness

Lemma For all assertions P , Q , and statements C :

$$\vdash \{P\}C\{Q\} \implies \models \{P\}C\{Q\}$$

Any triple derivable using the syntactic proof system holds semantically.